

Query Transformation Techniques

Retrieval quality depends on the query as much as on the index. Users write short, ambiguous questions; documents are written in longer, more formal language. Query transformation techniques rewrite, expand, or reshape the user's question before retrieval so that it matches the corpus better. They are applied at query time, require no re-indexing, and are often the cheapest way to lift retrieval quality.

Query rewriting

Query rewriting asks a language model to restate the user's question in a cleaner, more retrievable form: fixing typos, expanding abbreviations, resolving pronouns from conversation history, and removing filler words. In a chat setting this is essential: a follow-up like 'what about the second option?' retrieves nothing on its own, and must be condensed with the conversation history into a standalone question such as 'what are the drawbacks of HNSW indexing?'. This step is usually a single cheap model call before retrieval.

Multi-query expansion

Multi-query expansion generates several differently-worded variants of the question - typically three to five - and retrieves candidates for every variant. The result lists are then merged with Reciprocal Rank Fusion, exactly the same fusion used for hybrid dense-plus-sparse retrieval. Each variant probes the embedding space from a different angle, so passages that one phrasing misses are caught by another. The cost is one extra generation call plus one retrieval per variant, and the technique pairs naturally with any existing hybrid setup.

HyDE: Hypothetical Document Embeddings

HyDE flips the matching problem around. Questions and answers look different in embedding space: a short question may sit far from the long, factual passage that answers it. HyDE asks the model to write a hypothetical answer to the question - which may contain invented facts, and that is acceptable - then embeds that hypothetical document and uses ITS vector for the nearest-neighbour search. Because the fake answer is shaped like a real answer, document-to-document similarity is often higher than question-to-document similarity. The retrieved passages are real; only the search probe was synthetic.

Choosing among them

Rewriting is the default hygiene step and is nearly free. Multi-query is the reliable recall booster when queries are varied and short. HyDE shines when the vocabulary gap between questions and documents is large, such as technical corpora queried by non-experts. All three add at least one model call of latency before retrieval even begins, so measure the gain on a golden set before paying that cost in production.